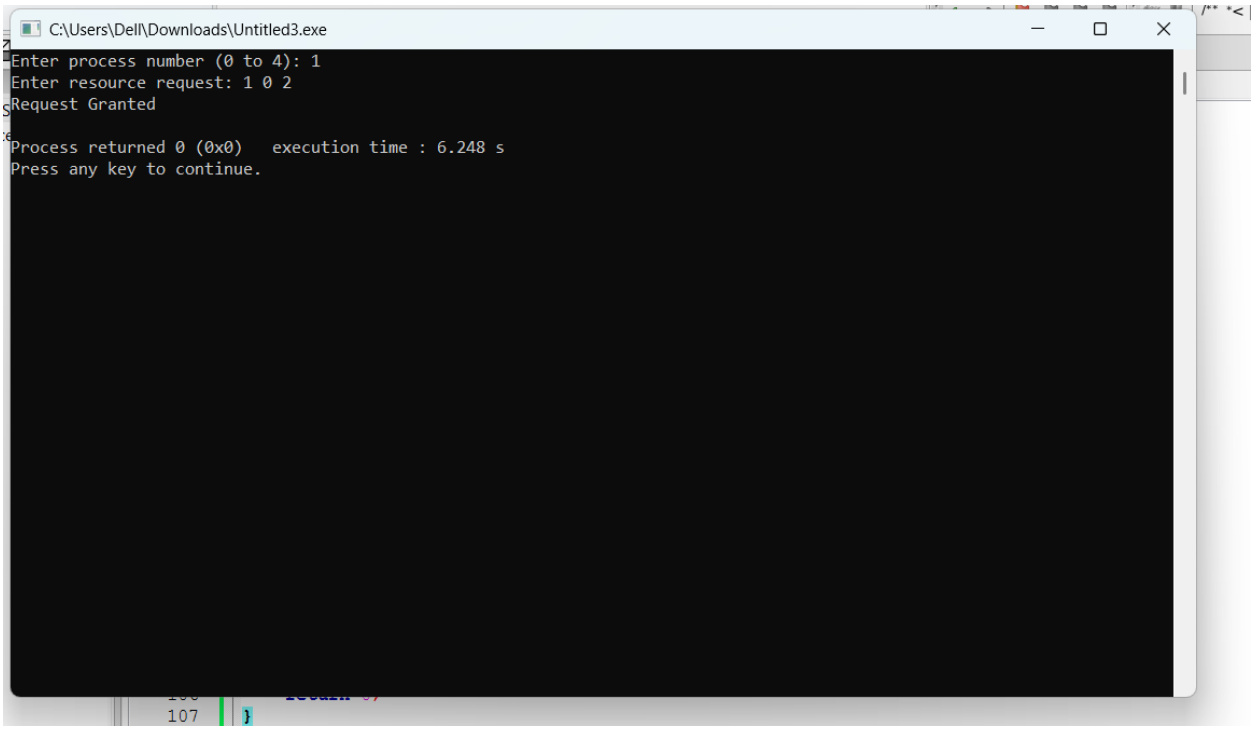
EXP 17

CODE:

```
#include <stdio.h>

#define MAX_PROCESSES 5
#define MAX_RESOURCES 3

int is_safe();

int available[MAX_RESOURCES] = {3, 3, 2};

int maximum[MAX_PROCESSES][MAX_RESOURCES] = {
   {7, 5, 3},
   {3, 2, 2},
   {9, 0, 2},
   {2, 2, 2},
   {4, 3, 3}
};

int allocation[MAX_PROCESSES][MAX_RESOURCES] = {
   {0, 1, 0},
   {2, 0, 0},
   {3, 0, 2},
   {2, 1, 1},
   {0, 0, 2}
};

int request_resources(int process_num, int request[]) {

   for (int i = 0; i < MAX_RESOURCES; i++) {
      if (request[i] > available[i] | |
         request[i] > (maximum[process_num][i] -
allocation[process_num][i]))
         return 0;
   }

   for (int i = 0; i < MAX_RESOURCES; i++) {
      available[i] -= request[i];
      allocation[process_num][i] += request[i];
   }

   if (is_safe()) {
      return 1;
   } else {
      for (int i = 0; i < MAX_RESOURCES; i++) {
         available[i] += request[i];
         allocation[process_num][i] -= request[i];
      }
      return 0;
   }
}
```

```
int is_safe() {
   int work[MAX_RESOURCES];
   int finish[MAX_PROCESSES] = {0};

   for (int i = 0; i < MAX_RESOURCES; i++) {
      work[i] = available[i];
   }

   int count = 0;

   while (count < MAX_PROCESSES) {
      int found = 0;

      for (int i = 0; i < MAX_PROCESSES; i++) {
         if (finish[i] == 0) {
            int j;
            for (j = 0; j < MAX_RESOURCES; j++) {
               if ((maximum[i][j] - allocation[i][j]) > work[j])
                  break;
            }

            if (j == MAX_RESOURCES) {
               for (int k = 0; k < MAX_RESOURCES; k++) {
                  work[k] += allocation[i][k];
               }
               finish[i] = 1;
               found = 1;
               count++;
            }
         }
      }

      if (found == 0)
         return 0;
   }

   return 1;
}

int main() {
   int process_num, request[MAX_RESOURCES];

   printf("Enter process number (0 to 4): ");
   scanf("%d", &process_num);

   printf("Enter resource request: ");
   for (int i = 0; i < MAX_RESOURCES; i++) {
      scanf("%d", &request[i]);
   }

   if (request_resources(process_num, request)) {
      printf("Request Granted\n");
   } else {
      printf("Request Denied\n");
   }

   return 0;
}
```

OUTPUT:


C:\Users\Dell\Downloads\Untitled3.exe
Enter process number (0 to 4): 1
Enter resource request: 1 0 2
Request Granted
Process returned 0 (0x0) execution time : 6.248 s
Press any key to continue.